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(54) **AIRCRAFT TURBOMACHINE COMPRISING
A DEVICE FOR LUBRICATING A BEARING**

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See application file for complete search history.

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(57) **ABSTRACT**

The invention relates to a turbomachine for an aircraft,
comprising:

a first rotor comprising a first shaft,

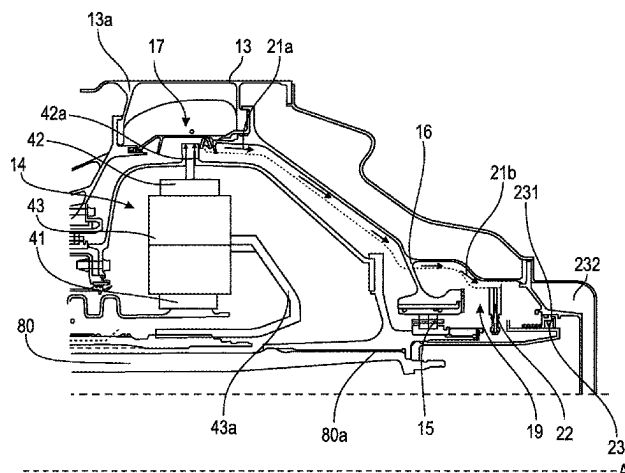
a second rotor comprising a second shaft,

a mechanical reduction gearing having an epicyclic gear
set comprising a sun gear connected to the second shaft,
a ring gear connected to the first shaft, and planet gears
located between the sun gear and the ring gear and
borne by a planet carrier attached to a stator of the
turbomachine,

rolling element bearings for guiding said first shaft and
second shaft in rotation,

an annular gutter which extends around the ring gear of
the reduction gearing and which is configured to

(Continued)



recover oil for lubricating the reduction gearing that is sprayed by centrifugal action out from the ring gear during operation, and
 an annular bearing support which is attached, with the gutter, to a stator of the turbomachine and which supports at least one of said bearings, characterized in that it also comprises:
 at least one device for conveying oil recovered by said gutter, which device is borne by said annular support and extends as far as said at least one bearing in order to lubricate the latter.

8 Claims, 3 Drawing Sheets

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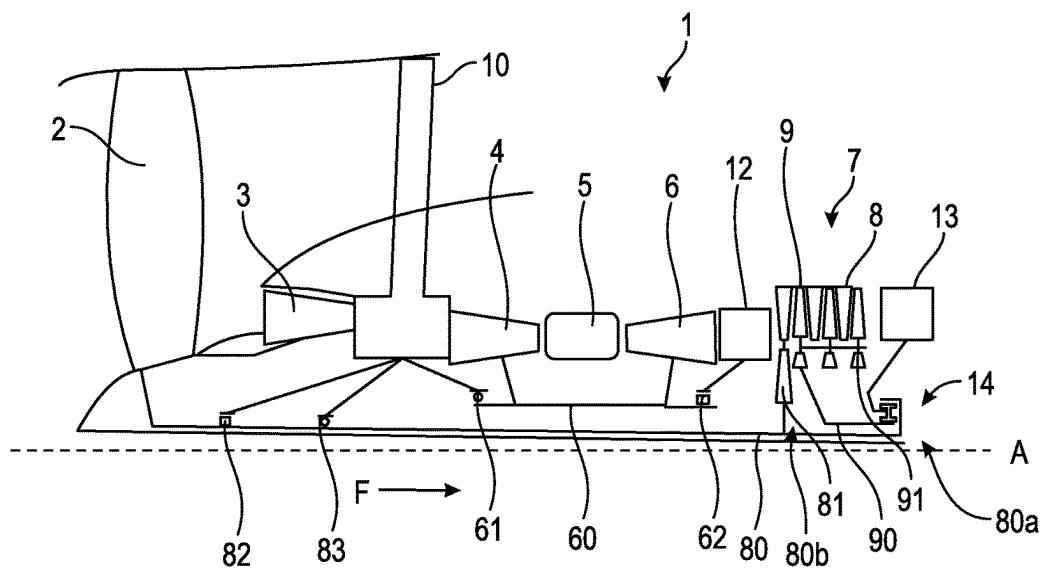


FIG. 1

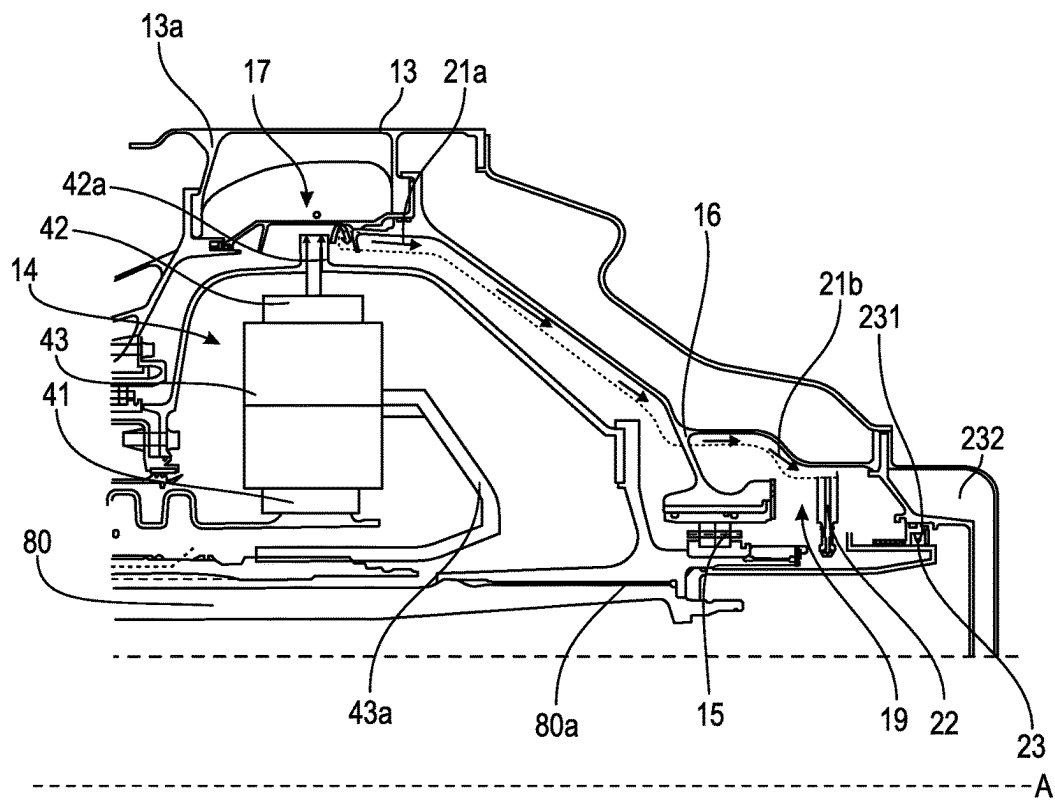


FIG. 2

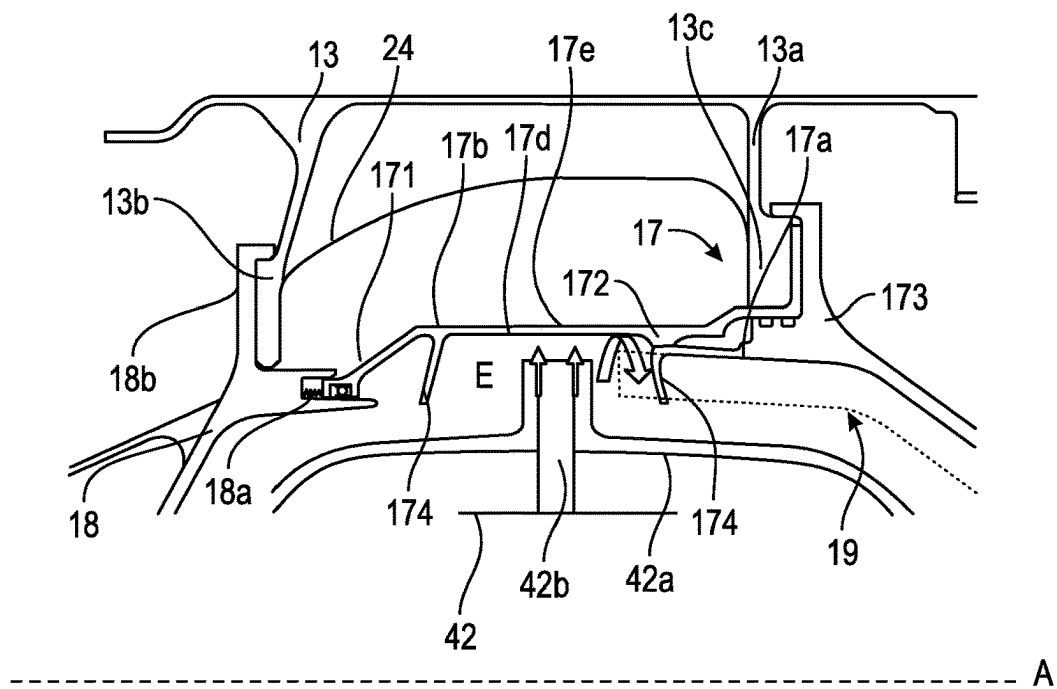


FIG. 3

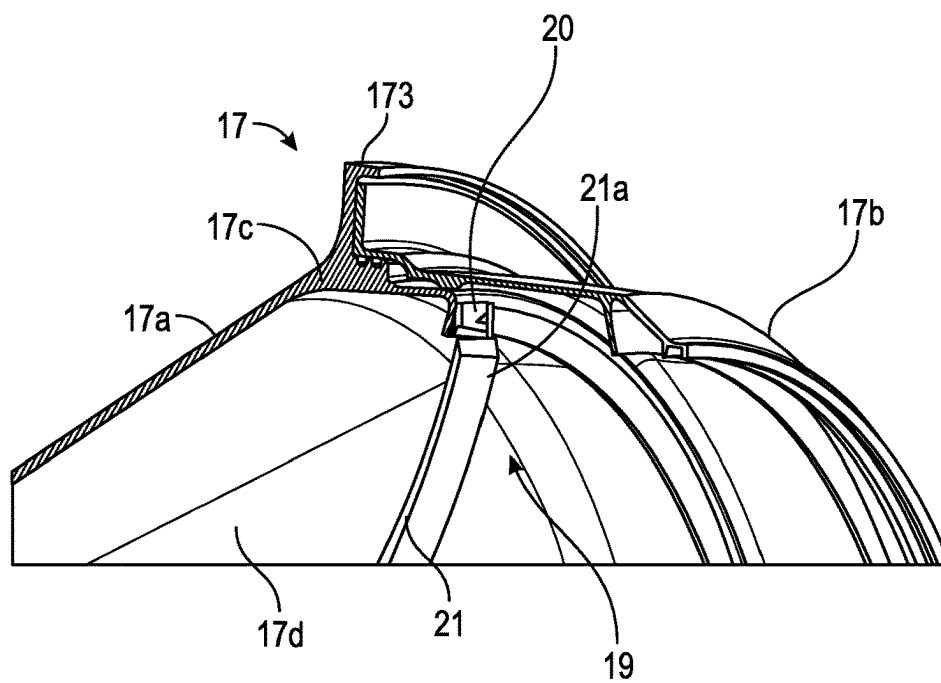


FIG. 4

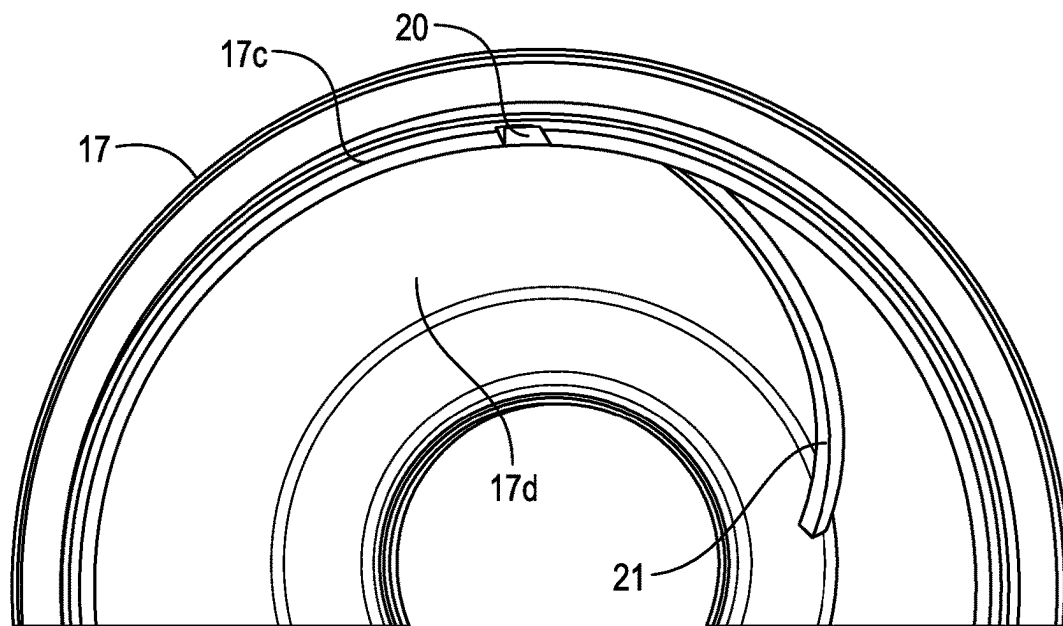


FIG. 5

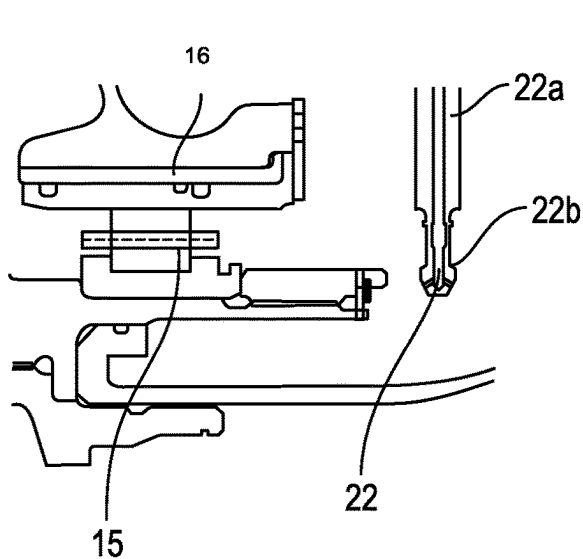


FIG. 6A

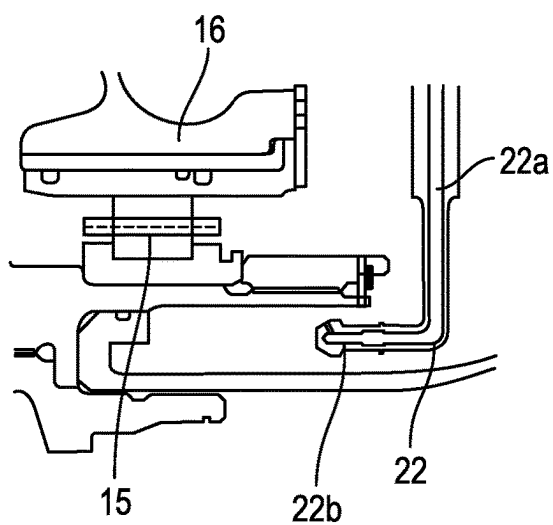


FIG. 6B

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AIRCRAFT TURBOMACHINE COMPRISING A DEVICE FOR LUBRICATING A BEARING

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a national phase application of International Patent Application PCT/FR2021/050790 ("the '790 application"), filed on May 9, 2021 and titled "AIRCRAFT TURBOMACHINE COMPRISING A DEVICE FOR LUBRICATING A BEARING," which application is related to and claims priority benefits of France Patent Application No. 2004717 ("the '717 application"), filed on May 13, 2020. The '790 and '717 applications are hereby incorporated in their entireties by this reference.

TECHNICAL FIELD OF THE INVENTION

The invention relates to the technical field of aircraft turbomachine.

More particularly, the invention falls within the field of aircraft turbomachine comprising a counter-rotating turbine.

TECHNICAL BACKGROUND

The prior art is illustrated by the documents EP-A1-3 575 562, US-A1-2015/361829, FR-A1-2 977 636 and EP-A1-3 460 199.

Conventionally, an aircraft turbomachine comprises from upstream to downstream, in the direction of gas flow, a fan, a low-pressure compressor, a high-pressure compressor, an annular combustion chamber, a high-pressure turbine and a low-pressure turbine. The rotor of the low-pressure compressor is driven by the rotor of the low-pressure turbine, and the rotor of the high-pressure compressor is driven by the rotor of the high-pressure turbine.

From an engine performance and fuel consumption point of view, it is advantageous to maximise the rotational speed of the low-pressure turbine as it allows to obtain a better efficiency of the turbine. However, increasing the rotational speed of the turbine means increasing the centrifugal forces on it, and therefore greatly complicates its design.

One suggestion for increasing the efficiency of a turbine without increasing its rotational speed is to use a counter-rotating turbine. The low-pressure turbine is replaced by a two-rotor turbine with a first rotor configured to rotate in a first direction of rotation and connected to a first turbine shaft, and a second rotor configured to rotate in an opposite direction of rotation and connected to a second turbine shaft. The first and second turbine shafts are centred and guided in rotation by a plurality of guide bearings. The first rotor has turbine wheels interposed between turbine wheels of the second rotor.

A low-pressure turbine may have a rotational speed at take-off of the order of 4,000 rpm in a conventional architecture where the turbine drives the fan directly or a rotational speed at take-off of the order of 10,000 rpm in an architecture where the turbine drives the fan through a reduction gear. Its replacement by a counter-rotating turbine, the rotors of which rotate respectively at take-off speeds of the order of 3,000 and 7,000 revolutions per minute makes it possible to have a relative speed of 10,000 revolutions per minute (3,000+7,000) while having an absolute speed in a low range of the above-mentioned speed interval.

This counter-rotating turbine thus comprises a slow rotor and a fast rotor, the slow rotor driving the fan and the fast rotor meshing with a planetary-type epicyclic reduction

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gear, the input and output of which are counter-rotating (rotating ring gear, fixed planet carrier, rotating sun gear).

The reduction gear couples the fast rotor to the slow rotor, allowing power to be transferred from the fast rotor to the slow rotor. This takes advantage of the higher efficiencies of a fast turbine while transferring a large part of the power from the turbine to the fan without passing through a reduction gear but through a shaft.

This architecture is complex because of its mechanical integration: the mechanical reduction gear is for example located downstream of the turbomachine, radially inside a stator annular casing called the exhaust casing or upstream of the turbomachine, radially inside a stator annular casing called the intermediate casing.

Furthermore, as the reduction gear releases considerable energy (of the order of 100 kW) during operation, it must be lubricated continuously in order to maintain an acceptable operating temperature. An oil circuit is therefore implemented to supply the reduction gear with oil. The discharge of the oil is an essential problem. Indeed, the amount of oil required to operate the reduction gear is significant and its discharge is therefore complex as it requires cumbersome discharging devices that are difficult to implement in such an environment.

In addition, the guide bearings in rotation of the first and second turbine shafts must also be lubricated with a lubricating oil. The conveyance of lubricating oil to the guide bearings is also a major problem. The amount of oil to be conveyed to the guide bearings is significant in this type of configuration. The conveying devices are therefore complex to implement in this type of environment, as are the oil discharging devices.

There is therefore a need to provide a turbomachine in which the oil for lubricating the reduction gear can be easily recovered and discharged and in which the bearings can be easily lubricated.

SUMMARY OF THE INVENTION

To this end, the invention provides a turbomachine for an aircraft, comprising:

- a first rotor comprising a first shaft,
- a second rotor comprising a second shaft,
- a mechanical epicyclic reduction gear comprising a sun gear connected to the second shaft, a ring gear connected to the first shaft, and planet gears located between the sun gear and the ring gear and carried by a planet carrier fixed to a stator of the turbomachine, rolling bearings for guiding in rotation of said first and second shafts,
- an annular gutter which extends around the ring gear of the reduction gear and which is configured to recover the oil for lubricating the reduction gear which is sprayed by centrifugal action out of the ring gear during operation, and
- an annular bearing support which is fixed with the gutter to a stator of the turbomachine and which supports at least one of said bearings.

The turbomachine is distinguished in that it further comprises at least one device for conveying oil recovered by said gutter, which is carried by said annular support and which extends to said at least one bearing for the purpose of lubricating it.

The turbomachine thus comprises a gutter for recovering oil for lubricating the reduction gear. According to the invention, the turbomachine also comprises a device for conveying at least a part of the oil recovered by the gutter

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which extends to the bearing. The conveying device thus allows the oil for lubricating the reduction gear recovered by the gutter to be used to lubricate the bearing. Thanks to the invention, it is therefore possible to dispense with a specific oil supply to lubricate the bearing, offering a considerable cost saving.

In addition, as a part of the oil for lubricating the reduction gear is used to lubricate the bearing, the total quantity of oil to be injected and discharged is reduced. The oil to be injected and discharged is conveyed through an oil circuit passing through arms of a casing of the turbomachine. It is therefore possible to reduce the size of the circuit, facilitating its integration into the arms of the casing of the turbomachine.

The turbomachine according to the invention may comprise one or more of the following features, taken in isolation from each other or in combination with each other:

- the conveying device comprises a scoop configured to take up the oil recovered by the gutter, the scoop being provided in an annular side wall of the gutter;
- the conveying device comprises at least one pipe for conveying oil taken up from the scoop to said at least one bearing, the pipe extending between a first end connected to the gutter and a second end extending to the bearing;
- the conveying pipe has a generally helical shape about a longitudinal axis of the turbomachine;
- the conveying pipe is formed in a rib projecting from an inner annular surface of said annular support, or by a tube carried by the annular support;
- the conveying device comprises at least one jet for spraying oil to said at least one bearing, the jet being carried by said annular support;
- said at least one jet has a generally rectilinear or angled shape;
- the first shaft is fixed to the ring gear and extends at least in part between the reduction gear and said annular support, this first shaft being guided by said at least one bearing supported by this annular support and lubricated by said device;
- said at least one jet is axially interposed between said at least one bearing and a dynamic annular seal mounted around the first shaft;
- an annular casing which extends at least in part around the reduction gear and to which said annular support and said gutter are fixed, this annular casing comprising arms extending radially with respect to a longitudinal axis of the turbomachine and passing through a flow vein of a gas flux inside the turbomachine.

BRIEF DESCRIPTION OF THE FIGURES

Further features and advantages will be apparent from the following description of a non-limiting embodiment of the invention with reference to the attached drawings in which:

FIG. 1 is a very schematic representation of an aircraft turbomachine half according to the invention;

FIG. 2 is an enlarged schematic representation of a part of the turbomachine of FIG. 1;

FIG. 3 is an enlarged view of a part of FIG. 2;

FIG. 4 is a perspective view of an element of the invention shown in FIG. 3;

FIG. 5 is a further perspective view of an element of the invention shown in FIG. 3;

FIG. 6a is a schematic representation of a first example of an embodiment of a jet that may be fitted to the invention;

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FIG. 6b is a schematic representation of a second example of an embodiment of a jet that can be fitted to the invention;

DETAILED DESCRIPTION OF THE INVENTION

An aircraft turbomachine 1 is for example shown in FIG. 1. The turbomachine 1 is, for example, a counter-rotating turbine for turbomachine 1. The turbomachine 1 extends along a longitudinal axis A.

The turbomachine 1 comprises, from upstream to downstream in the direction of gas flow F, a fan 2, a low-pressure compressor 3, a high-pressure compressor 4, an annular combustion chamber 5, a high-pressure turbine 6 and a counter-rotating turbine 7.

The turbomachine 1 further comprises a plurality of structural casings. For example, it comprises an intermediate casing 10 arranged between the low-pressure compressor 3 and the high-pressure compressor 4. The turbomachine 1 further comprises a turbine casing 12. The turbine casing 12 is for example equipped with arms forming rectifier vanes and is called TVF (Turbine Vane Frame). The turbine casing 12 is arranged between the high-pressure turbine 6 and the counter-rotating turbine 7. The turbomachine 1 also includes an exhaust casing 13. The exhaust casing 13 is for example the last turbine casing and is called TRF (Turbine Rear Frame).

The high-pressure turbine 6 comprises a rotor which drives in rotation a rotor of the high-pressure compressor 4 via a high-pressure shaft 60. The high-pressure shaft 60 is centred and guided in rotation by high-pressure bearings, such as an upstream high-pressure bearing 61 and a downstream high-pressure bearing 62. The upstream high-pressure bearing 61 is for example a ball bearing and the downstream high-pressure bearing 62 is for example a roller bearing. The upstream high pressure bearing 61 is mounted between an upstream end of the high-pressure shaft 60 and the intermediate casing 10.

The counter-rotating turbine 7 comprises a first rotor 8 comprising a first shaft 80 and a second rotor 9 comprising a second shaft 90.

More particularly, the first rotor 8 comprises a first series of wheels 81 configured to rotate in a first direction of rotation and connected to the first shaft 80. The second rotor 9 comprises a second series of wheels 91 configured to rotate in a direction opposite to the first direction of rotation and connected to the second shaft 90. The wheels 91 of the second series are interposed between the wheels 81 of the first series.

Each wheel 81, 91 comprises an annular row of blades each comprising an aerodynamic profile having a pressure side and suction side which meet to form a leading edge and a trailing edge of the gases in a vein of the counter-rotating turbine 7.

The first shaft 80 drives in rotation the fan 2 and the rotor of the low-pressure compressor 3.

In addition, the turbomachine 1 comprises rolling bearings for rotationally guiding the first shaft 80 and second shaft 90.

According to an example not shown, the second shaft 90 is centred and guided in rotation by means of two upstream and downstream guiding rolling bearings.

The first shaft 80 is centred and rotationally guided downstream by at least one guide bearing, for example by at least one downstream guide bearing 15 visible in FIG. 2.

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The first shaft **80** is centred and rotationally guided upstream by two upstream guide bearings **82**, **83** mounted between the upstream end of the first shaft **80** and the intermediate casing **10**.

In addition, the turbomachine **1** includes a mechanical epicyclic reduction gear **14**. As best seen in FIG. 2, the reduction gear **14** comprises a sun gear **41** connected to the second shaft **90**, a ring gear **42** connected to the first shaft **80** and planet gears **43** located between the sun gear **41** and the ring gear **42** and carried by a planet carrier **43a** fixed to a stator of the turbomachine **1**.

More particularly, the first shaft **80** has a downstream end **80a** attached to an upstream end of a ring gear carrier **42a** of the ring gear **42** and an upstream end **80b** attached to the last stage of the first rotor **81** of the first turbine **8** as shown in FIG. 1.

The turbine casing **12** and the exhaust casing **13** comprise a central hub, and an outer ring which surrounds the hub and is connected thereto by a series of arms **13a**. The arms **13a** are substantially radial or inclined to the longitudinal axis **A** of the turbomachine **1** and pass through the vein of the turbine. The central hub of the exhaust casing **13** extends around at least part of the reduction gear **14**.

The downstream bearing **15** is located downstream of the reduction gear **14**. The downstream bearing **15** is supported by an annular bearing support **16** attached to a stator of the turbomachine **1**, for example to the exhaust casing **13**. The first shaft **80** extends at least partially between the reduction gear **14** and the support **16**.

Furthermore, an annular dynamic seal **23** is mounted around the first shaft **80**. The dynamic seal **23** is for example a segmented radial seal (JRS). The dynamic seal **23** is for example made of carbon. The dynamic seal **23** is, for example, housed in a groove **231** provided in a flange **232** of a fixed structure connected to the stator of the turbomachine **1**, for example to the exhaust casing **13** and the support **16**.

In operation, the reduction gear **14** is lubricated and the oil that has lubricated the reduction gear must be recovered and discharged for recycling to prevent this oil from accumulating in the reduction gear **14** and becoming coke under the effect of the high temperatures.

To this end, the turbomachine **1** comprises an annular gutter **17** which extends around the ring gear **42** of the reduction gear **14** and which is configured to recover the oil for lubricating the reduction gear **14** which is sprayed centrifugally out of the ring gear **42** during operation as illustrated by the rounded arrow in FIG. 2.

Specifically, the oil is centrifuged and ends up in the ring gear **42** of the reduction gear **14**. As best seen in FIG. 3, passages **42b** may be provided at the outer periphery of the ring gear **42** so as to centrifugally spray the oil onto the gutter **17** extending opposite these passages **42b**. The passages **42b** are, for example, oriented radially with respect to the longitudinal axis **A** of the turbomachine **1**.

The gutter **17** is fixed with the annular bearing support **16** to the stator of the turbomachine **1**, for example to the exhaust casing **13**.

As best seen in FIGS. 4 and 5, the gutter **17** has an annular side wall **17c**. The gutter **17** further has an inner annular surface **17d** and an outer annular surface **17e**.

The gutter **17** is formed, for example, by the assembly of a first annular wall **17a** and a second annular wall **17b** mounted coaxially one inside the other. The first annular wall **17a** is for example formed by the support **16**.

Each of the walls **17a**, **17b** includes an annular discharge rib **174** radially projecting towards the inside.

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The annular discharge ribs **174** axially delimit between them an oil recovery space **E**. The space **E** is radially delimited by the inner annular surface **17d**.

In addition, the outer annular surface **17e** is advantageously coated with an annular thermal insulation coating **24**. The annular coating **24** is thus arranged radially between the exhaust casing **13** and the gutter **17**. The annular lining **24** is separated from the exhaust casing **13** by an annular free space for air circulation and ventilation for example.

The second wall **17b** comprises an upstream end **171** and a downstream end **172**. The upstream end **171** of the second wall **17b** is axially engaged in an annular groove **18a** of a member **18** fixed to an upstream flange **13b** of the exhaust casing **13** via a fixing flange **18b**. The upstream flange **13b** of the exhaust casing **13** is located substantially in line with the arms **13a** of the exhaust casing **13**. Preferably, an annular seal is arranged in the groove **13a** and is in axial abutment with the upstream end **171** of the second wall **17b**.

The downstream end **172** of the second wall **17b** comprises a flange fixed to a downstream flange **13c** of the exhaust casing **13**, located substantially in line with the arms **13a**.

The first wall **17a** comprises a flange fixed to the downstream flange **13c** of the exhaust casing **13**.

The fixing flange forms a cylindrical rim **173** facing upstream to cooperate with the second wall **17b**.

The gutter **17** includes openings (not shown) which ensure the passage of oil to discharge ducts (not shown). The ducts pass through the arms **13a** of the exhaust casing **28**. Advantageously, the gutter **17** comprises an oil recovery chamber (not shown) located at the lower part. The recovery chamber provides for the tranquillisation of the oil, i.e. part of its degassing or deaeration in order to reduce the number of air bubbles in the oil. Thus, the oil passes through the openings that communicate with the recovery chamber before being discharged through the ducts through the arms **13a** of the exhaust casing **13** for example.

In order to reduce the total amount of oil to be conveyed and the amount of oil to be discharged through the arms **13a**, and thus reduce the bulk of the ducts passing through the arms **13a**, according to the invention, the turbomachine further comprises at least one device **19** for conveying oil recovered by the gutter **17** carried by the annular support **16** and which extends to at least one guide bearing, for example to the downstream guide bearing **15** for lubrication thereof.

As shown in FIG. 4, the device **19** comprises, for example, a scoop **20** for drawing off oil, at least one pipe **21** for conveying oil and at least one jet **22** (FIG. 2) for spraying oil to the guide bearing such as the downstream guide bearing **15** in rotation of the first shaft **80**.

The scoop **20** is configured to take up oil collected by the gutter **17**. The scoop **20** is provided in the annular side wall **17c** of the gutter **17**, for example of the first wall **17a**.

The scoop **20** is in fluid communication with the pipe **21**. The pipe **21** is used to convey the oil taken by the scoop **20** to the guide bearing, for example the downstream guide bearing **15**.

As shown for example in FIG. 5, the pipe **21** has a helical shape around the longitudinal axis **A** of the turbomachine **1**. The helical shape of the pipe **21** allows to limit the pressure losses of the oil taken by the scoop **20** during its conveyance to the guide bearing. The pipe **21** extends between a first end **21a** connected to the gutter **17** and a second end **21b** extending to the downstream bearing **15**. The second end **21b** is connected to the jet **22**. More particularly, the first end **21a** is connected to the first wall **17a** of the gutter **17** and the

second end **21b** is connected to the jet **22**, the pipe **21** extending along the support **16** between the first end **21a** and the second end **21b**.

In the example shown in FIG. 5, the pipe **21** is formed in a projecting rib on the inner annular surface **17d** of the support **16**. For example, the pipe **21** is formed in the rib **174** of the first wall **17a** of the gutter **17** and extends into the support **16**.

According to another example not shown, the pipe **21** is formed by a tube carried by the support **16**.

The jet **22** is used to distribute the oil conveyed by the pipe **21** to the bearing **15**. The jet **22** is for example carried by the support **16**. The jet **22** is fixed to a downstream end of the support **16**. The jet **22** is interposed axially between the bearing **15** and the dynamic seal **23**. In order to limit the oil spraying on the dynamic seal **23**, a spinner may be placed against the dynamic seal **23**.

More particularly, the jet **22** comprises a cylindrical body **22a** and an ejection nozzle **22b**.

According to a first example embodiment shown in FIG. 6a, the jet **22** has a generally rectilinear shape. Thus, the cylindrical body **22a** and the ejection nozzle **22b** extend along the same axis perpendicular to the axis of the turbomachine **1**. According to another embodiment shown in FIG. 6b, the jet **22** has a generally angled shape. The ejection nozzle **22b** thus extends radially with respect to the cylindrical body **22a** such that the ejection nozzle **22b** extends parallel to the longitudinal axis A of the turbomachine **1**. In this example, the ejection nozzle **22b** is oriented towards the bearing **15**. This configuration makes it possible to maximise the quantity of oil distributed to the bearing **15** in particular in the event of a loss of pressure during the conveyance of the oil in the pipe **21** resulting in too low a pressure in the jet **22**.

The invention thus implements the gutter **17**, which allows to recover the oil for lubricating the reduction gear **14** in order to limit the risks of coke formation in the reduction gear **14**. Part of this oil is conveyed by the device **19** to the guide bearing, for example to the downstream guide bearing **15** in rotation of the first shaft **80** to lubricate it. In particular, the part of the oil conveyed is first taken from the gutter **17** by the scoop **20** which communicates with the pipe **21** allowing the oil to be conveyed to the jet **23** distributing the oil taken to the downstream guide bearing **15**. As a result, the amount of oil remaining in the gutter **17** and discharged is less. The ducts of the gutter **17** extending into the arms **13a** of the exhaust casing **13** can thus be sized to discharge a smaller amount of oil. This results in smaller ducts that can be integrated into the exhaust casing **13**. Also, the costs associated with lubricating the reduction gear **14** and the bearing are reduced since the total amount of oil introduced is reduced by reusing some of the oil for lubricating the reduction gear **14**.

Furthermore, the invention has been described in relation to a turbomachine **1** in which the reduction gear **14** is located downstream of the turbomachine **1** and is surrounded by the exhaust casing **13**. The invention also applies to a turbomachine in which the reduction gear **14** is located upstream of the turbomachine **1** and is surrounded by the intermediate casing **10**. According to this embodiment, the gutter **17** and the support **16** are attached to the intermediate casing **10**. The same advantages result from this embodiment.

The invention claimed is:

1. A turbomachine for an aircraft, comprising:
 - a first rotor comprising a first shaft,
 - a second rotor comprising a second shaft,
 - a mechanical epicyclic reduction gear comprising a sun gear connected to the second shaft, a ring gear connected to the first shaft, and planet gears located between the sun gear and the ring gear and carried by a planet carrier,
 - rolling bearings for guiding in rotation said first and second shafts,
 - an annular gutter which extends around the ring gear of the reduction gear and which is configured to recover for lubricating the reduction gear which is sprayed by centrifugal action out of the ring gear during operation, and
 - an annular bearing support which is fixed with the gutter to a stator of the turbomachine or to another stator of the turbomachine and which supports at least one of said bearings,
- wherein the turbomachine further comprises:
 - at least one device for conveying the oil recovered by said gutter, which is carried by said annular support and which extends to said at least one bearing with a view to lubricating the at least one bearing,
 - the conveying device comprising:
 - a scoop configured to take up the oil recovered by the gutter, the scoop being provided in an annular side wall of the gutter, and
 - at least one pipe for conveying the oil taken up by the scoop to said at least one bearing, the pipe extending between a first end connected to the gutter and a second end extending to the bearing.
2. The turbomachine according to claim 1, wherein the conveying pipe has a generally helical shape about a longitudinal axis of the turbomachine.
3. The turbomachine according to claim 1, wherein the conveying pipe is formed:
 - in a rib projecting from an inner annular surface of said annular support, or
 - by a tube carried by the annular support.
4. The turbomachine according to claim 1, wherein the conveying device comprises at least one jet for spraying oil to said at least one bearing, the jet being carried by said annular support.
5. The turbomachine according to claim 4, wherein said at least one jet has a generally rectilinear or angled shape.
6. The turbomachine according to claim 5, wherein said at least one jet is interposed axially between said at least one bearing and a dynamic annular seal mounted around the first shaft.
7. The turbomachine according to claim 1, wherein the first shaft is fixed to the ring gear and extends at least in part between the reduction gear and said annular support, this first shaft being guided by said at least one bearing supported by this annular support and lubricated by said device.
8. The turbomachine according to claim 1, further comprising an annular casing which extends at least in part around the reduction gear and to which said annular support and said gutter are fixed, the annular casing comprising arms extending radially with respect to a longitudinal axis of the turbomachine and passing through a flow vein of a gas flux inside the turbomachine.